

SE4095 — Social Network: Theory and Practice

MASTER REVISION

Master Exam Revision

Every definition, formula, and comparison in one place — plus cross-topic scenario practice that trains you to pick the right tool under exam pressure.

Synthesizes Parts 1–3. Use this for final review; use the three Parts when you need the full "why" behind any one entry.

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Important Definitions

Term	Definition
Social Network Analysis (SNA)	Visualizing/analyzing relationships and connections between entities — studies relationships, not just individuals.
Homophily / Heterophily	Similar nodes tend to connect ("birds of a feather") / diverse nodes connect ("opposites attract").
Degree / Neighborhood	d_i = number of edges touching node i ; $N(v)$ = the set of nodes v shares an edge with.
Walk / Trail / Path	Walk: repeats allowed on edges & nodes. Trail: edges distinct. Path: edges & nodes distinct. Closed versions: closed walk / tour / cycle.
Connected / Strongly / Weakly connected	A path exists between every pair (undirected) / a directed path exists both ways for every pair / only connected once direction is dropped.
Diameter	The length of the <i>longest</i> shortest path between any pair of nodes in the graph.
Bridge / Local bridge	An edge whose removal increases component count (rare in practice) / an edge between two nodes with no common friends, whose removal pushes their distance past 2 (the realistic version; has a "span").
Strong Triadic Closure	If u has strong ties to both v and w , but $v-w$ have no tie at all, u violates the property.
Centrality (general)	How important a node is within a network — either via its connections (degree/eigenvector/Katz/PageRank) or as a broker (betweenness/closeness).
Transitivity / Reciprocity	A friend of my friend is my friend / a 2-node special case: mutual ties ("if you friend me, I'll friend you").
Social balance	A signed triangle is balanced iff the product of its 3 edge signs is positive.
Structural equivalence	Two nodes are similar to the degree that they share the same neighborhood.
Community	A set of ≥ 2 nodes sharing an interest, with interactions with respect to it — dense internal ties, sparse external ties.
Explicit (emic) / Implicit (etic) community	Formed by direct subscription / formed purely from interaction patterns with no formal joining.
Clique	A maximum complete subgraph — every member directly connects to every other member.
Modularity (Q)	A measure of community-division quality: actual within-community edges minus the number expected at random.
Purity / NMI	Fraction of nodes matching their community's majority label (gameable by singletons) / information-theoretic agreement between found communities and true labels, bounded $[0,1]$.

Important Formulas

Degree centrality

$$C_d(v_i) = d_i, \text{ normalized: } d_i/(n-1)$$

Eigenvector centrality

$$\lambda \mathbf{C}_e = A^T \mathbf{C}_e - \text{take largest } \lambda$$

Katz centrality

$$\mathbf{C}_{Katz} = \beta(I - \alpha A^T)^{-1} \cdot \mathbf{1}, \alpha < 1/\lambda_{\max}$$

Betweenness centrality

$$C_b(v_i) = \sum \sigma_{st}(v_i) / \sigma_{st}$$

Closeness centrality

$$C_c(v_i) = 1 / \left(\frac{1}{n-1} \sum_j l_{i,j} \right)$$

Reciprocity

$$R = \frac{1}{m} \text{Tr}(A^2)$$

Jaccard / Cosine similarity

$$\frac{|N_i \cap N_j|}{|N_i \cup N_j|} / \frac{|N_i \cap N_j|}{\sqrt{|N_i||N_j|}}$$

Expected similarity overlap

$$d_i d_j / n \text{ (random-chance baseline)}$$

Modularity

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

Precision / Recall / F

$$\frac{TP}{TP+FP}, \frac{TP}{TP+FN}, \frac{2PR}{P+R}$$

Degree sum theorem

$$\sum_i d_i = 2|E| \text{ (undirected)}$$

Tree property

$$|V| = |E| + 1$$

Important Comparisons

Concept	Main Purpose	Works Best When	Key Idea	Limitation	Exam Clue
Degree centrality	Local popularity	Quick first pass	Count direct ties	Blind to rest of network	"most friends"
Eigenvector centrality	Importance via important neighbors	Undirected, well-connected	Weight ties by neighbor's score	Can be exactly 0 (directed/DAG)	"connected to important people"
Katz centrality	Fixes eigenvector's zero problem	Directed graphs, DAGs	Add flat baseline β	Over-credits celebrities	"no node stuck at zero"
PageRank	Fixes Katz's celebrity problem	Web/social directed graphs	Split centrality across out-links	Needs dangling-node handling	"not everyone a celebrity follows is famous"
Betweenness	Brokerage	Finding bottlenecks	Fraction of shortest paths through node	Expensive: $O(nm)$	"connects two groups"
Closeness	Speed of reach	Fastest info spread	Inverse avg. distance	Breaks on disconnected graphs	"reaches everyone quickly"
Clique / k-plex / k-core	Tight-knit member-based groups	Groups must be near-complete	Direct/near-direct connections	NP-hard, rare, fragile	"everyone knows everyone"
CPM	Overlapping communities	Nodes belong to >1 group	Chain overlapping cliques	Sensitive to k choice	"a person belongs to multiple groups"
Modularity Max. (Louvain)	Global best partition	General-purpose, scalable	Maximize internal-expected density	Local optimum only, resolution limit	"maximize internal vs external density"
Girvan-Newman	Hierarchical, divisive detection	Want the full dendrogram	Remove highest edge-betweenness first	Very expensive	"remove the edges connecting groups"
SBM	Probabilistic block inference	Ties are probabilistic, not certain	Infer hidden blocks from connections	Needs a generative assumption	"infer hidden structure from connections"
Label Propagation	Fast, simple, scalable detection	Huge, dynamic networks	Adopt majority neighbor label	Can be unstable/non-deterministic	"must scale, simplicity matters"

Concept Selection Guide

If the question says...	Think about...	Why
"most direct connections / most friends"	Degree centrality	Literal definition of degree
"important because connected to important people"	Eigenvector centrality (or Katz/PageRank if directed)	Weights neighbor importance, not just count
"directed graph, some nodes might get zero score unfairly"	Katz centrality	Bias term β guarantees no one is stuck at zero
"celebrity shouldn't get full credit for every follow"	PageRank	Splits centrality across out-links instead of copying it fully
"connects two otherwise separate groups / bottleneck"	Betweenness centrality	Measures fraction of shortest paths through the node
"can reach everyone quickly / fastest spreader"	Closeness centrality	Inverse of average shortest-path distance
"visit every edge exactly once"	Euler's condition (degree parity)	Existence check, not a search algorithm
"shortest path, unweighted graph"	BFS	Level-order guarantees shortest path on first visit
"shortest path, weighted, non-negative"	Dijkstra's algorithm	Purpose-built for weighted shortest paths
"connect everything as cheaply as possible"	Prim's algorithm (MST)	Minimizes total edge weight while spanning every node
"a friend of my friend is my friend"	Transitivity	Direct definition
"if you friend me, I'll friend you"	Reciprocity	2-node mutual-tie special case
"friend/enemy triangle, is it stable?"	Social Balance Theory	Product of 3 signs; positive = balanced
"who outranks whom, is it consistent?"	Social Status Theory	Directed version of balance; check for chained contradictions
"do these two nodes have the same neighbors?"	Structural equivalence (Jaccard/Cosine)	Similarity by shared neighborhood, not attributes
"are two popular nodes' overlaps meaningfully similar, or just because both are popular?"	Similarity significance	Compare against expected random overlap $d_i d_j / n$
"members must all directly know each other"	Clique / k-plex / k-core	Near-completeness requirement
"a person belongs to multiple communities"	Clique Percolation Method (CPM)	Only common method producing overlapping output

<i>"maximize internal density vs external sparsity," general case</i>	Modularity Maximization (Louvain)	Literal definition of modularity; scalable
<i>"nested / multi-level community structure"</i>	Hierarchical Clustering or Girvan–Newman	Both produce a dendrogram
<i>"ties are probabilistic, infer hidden groups"</i>	Stochastic Block Model	Purpose-built for probabilistic block inference
<i>"huge, fast-changing network, need speed"</i>	Label Propagation	Cheapest, fastest, simplest
<i>"where does information/flow get stuck?"</i>	InfoMap	Models random-walker dwell time
<i>"community grew, shrank, split, merged, appeared, or vanished"</i>	Community Evolution vocabulary	Map directly onto growth/contraction/split/merge/birth/death
<i>"we have correct labels to check against"</i>	Precision/Recall/F, Purity, or NMI	Ground-truth evaluation family
<i>"no labels available at all"</i>	Semantic coherence + modularity/conductance	The only options without ground truth — use both together
<i>"soft/probabilistic community membership needed"</i>	Gaussian Mixture Models (GMM)	Only method giving explicit membership probabilities
<i>"some node labels already known, rest unknown"</i>	Graph Convolutional Networks (GCN)	The semi-supervised option

EXAM TIP — the meta-skill this table is training

Nearly every scenario question in this course is secretly asking: "which of these two or three closely related tools is the precise fit, and why do the others fall slightly short?" Always name the alternative you're rejecting and the specific reason — that's usually worth as many marks as naming the right answer.

Scenario Practice — Full Syllabus

Each scenario below follows the same structure: identify the concept, explain why it fits, explain why the tempting alternative is a worse fit, then give the exact reasoning to write in an exam answer.

Scenario 1 — Foundations

"A university wants to understand which students, if given a fake news article, would cause it to spread fastest through the student body."

- 1. Concept:** Closeness centrality (or degree as a weaker first pass).
- 2. Why it applies:** "Spread fastest" is about speed of reach — closeness directly measures how quickly a node can reach the rest of the network on average.
- 3. Why the alternative falls short:** Betweenness measures brokerage/control over flow between others, not how fast a node itself can broadcast outward — a high-betweenness node might sit on a bottleneck between two clusters without being close to *everyone*.
- 4. Exam-ready reasoning:** "Closeness centrality is defined as the inverse of average shortest-path distance to all other nodes, which directly operationalizes 'speed of reach' — the node with highest closeness can broadcast to the whole network in the fewest average hops."

Scenario 2 — Graph Essentials

"A logistics company wants to lay fiber-optic cable connecting all 40 of its warehouses at minimum total cost, using known point-to-point trenching costs."

- 1. Concept:** Prim's algorithm (Minimum Spanning Tree).
- 2. Why it applies:** The goal is connecting every node at minimum *total* weight — the exact definition of an MST.
- 3. Why the alternative falls short:** Dijkstra's finds the cheapest path between two specific points, not the cheapest way to connect all 40 simultaneously — using it warehouse-pair by warehouse-pair would not minimize total cable used and could re-lay redundant connections.
- 4. Exam-ready reasoning:** "This is an MST problem, not a shortest-path problem, because the requirement is to connect every node at minimum total edge weight, not to minimize the distance between two particular endpoints. Prim's algorithm greedily grows a tree by always adding the cheapest edge on the current frontier, which guarantees a minimum spanning tree."

Scenario 3 — Centrality

"On a citation network (papers citing papers), a paper has very few citations but is cited by two or three extremely influential, highly-cited papers."

- 1. Concept:** Eigenvector centrality (or Katz, since citation graphs are directed and acyclic-ish).
- 2. Why it applies:** The paper's importance comes from *who* cites it, not how many citations it has — exactly the distinction eigenvector centrality is built to capture.
- 3. Why the alternative falls short:** Degree centrality would rank this paper low, since raw citation count is small — it can't see that its few citations are exceptionally high-value.
- 4. Exam-ready reasoning:** "Degree centrality treats every citation equally, so it undervalues this paper. Eigenvector centrality weights each incoming citation by the citing paper's own centrality, so being cited by a few highly central papers can outweigh being cited by many unimportant ones — this matches the observed pattern directly."

Scenario 4 — Friendship Patterns

"In an office network, employee A trusts B highly, and B trusts C highly, but A explicitly distrusts C."

- 1. Concept:** Social Balance Theory (if trust/distrust is treated as an undirected signed relationship) — or Status Theory if trust is read as a directed ranking.
- 2. Why it applies:** This is a signed triangle (+,+,−), and balance theory directly classifies triangles as stable or unstable from their sign product.
- 3. Why the alternative falls short:** Reciprocity only concerns 2-node mutual ties and says nothing about 3-node consistency; transitivity concerns friend-chains, not signed friend/enemy consistency.
- 4. Exam-ready reasoning:** "The sign product is (+)(+)(−) = negative, so this triangle is unbalanced. Balance theory predicts this configuration is socially unstable and under pressure to resolve — either A comes to trust C, or one of the existing positive ties breaks down."

Scenario 5 — Similarity

"Two YouTube channels each have 2 million subscribers. Analysis shows 50,000 subscribers follow both channels. Is this a meaningful overlap?"

- 1. Concept:** Similarity significance (expected overlap under randomness), not raw Jaccard/cosine alone.
- 2. Why it applies:** Both nodes have huge degree, so *some* overlap is expected by chance alone — the question is whether 50,000 is more than that baseline.
- 3. Why the alternative falls short:** Reporting raw Jaccard or cosine similarity alone doesn't tell you whether the overlap is surprising; two high-degree nodes can look "similar" by raw overlap count while actually being less similar than chance would predict, once you normalize.
- 4. Exam-ready reasoning:** "Compute the expected overlap under random neighbor selection, $d_i d_j / n$. If the actual 50,000 is well above this expected value, the channels share a meaningfully overlapping audience; if it's close to or below the expected value, the overlap is just what you'd predict from both channels' sheer popularity, not genuine audience similarity."

Scenario 6 — Community Detection (member-based)

"A co-authorship network shows several triples of researchers who all wrote a paper together, and some of these triples share two authors with a neighboring triple."

1. Concept: Clique Percolation Method (CPM).

2. Why it applies: The scenario describes exactly CPM's mechanism: $k=3$ cliques (co-authored triples) that share $k-1=2$ members with a neighbor, which CPM explicitly chains into one community.

3. Why the alternative falls short: Plain clique detection would treat each triple as an isolated 3-node community and miss that researchers spanning two triples should connect those groups into one larger (possibly overlapping) research community.

4. Exam-ready reasoning: "Since research groups naturally overlap (a researcher can belong to multiple collaboration triples), CPM is the correct choice — it builds a clique graph where cliques sharing $k-1$ nodes are linked, and connected components of that clique graph become the final (possibly overlapping) communities."

Scenario 7 — Community Detection (group-based)

"A social platform with 500 million users wants to detect communities in near real-time as new connections form, and can tolerate an approximate answer."

1. Concept: Label Propagation.

2. Why it applies: The requirements (huge scale, near real-time, approximate answer acceptable) map directly onto label propagation's strengths: it's the cheapest, fastest, simplest algorithm on the syllabus.

3. Why the alternative falls short: Girvan–Newman is explicitly too expensive at this scale (recomputing betweenness after every edge removal); Louvain is more scalable than Girvan–Newman but still costlier per-update than label propagation's simple neighbor-majority rule, which is the better fit when updates need to be near-instant.

4. Exam-ready reasoning: "Label propagation requires no global optimization step — each node simply adopts its neighbors' most common label — making it far cheaper to re-run incrementally as the graph changes, which is essential at this scale and update frequency."

Scenario 8 — Community Evolution & Evaluation

"A researcher detects communities in a network monthly for a year, and in month 7, one large community appears to have split into two topic-specific groups. They want to check whether this detected split is 'real' but have no labeled ground truth for month 7."

1. Concepts: Splitting (evolution vocabulary) + evaluation without ground truth (semantic coherence and/or modularity/conductance).

2. Why they apply: "One community becomes two" is definitionally splitting, not death (the members didn't vanish) or contraction (it's not just shrinking, it's dividing). With no ground truth, the only valid evaluation route is semantic coherence (do the two new communities' content/keywords form distinct, coherent topics) plus a structural quality measure like modularity or conductance.

3. Why the alternative falls short: Reaching for Precision/Recall/F/Purity/NMI here is a trap — all of them require labeled ground truth, which this scenario explicitly says is unavailable.

4. Exam-ready reasoning: "This is a splitting event. Since no ground truth exists for month 7, validate it without ground truth: check whether each resulting community's top keywords form a coherent, distinct topic, and confirm the split improved (or at least didn't harm) modularity — using both together is more reliable than either alone."

Scenario 9 — Cross-topic integration

"A network analyst has: (a) a node with very high betweenness but low degree, and (b) has just detected that this node's community lost several peripheral members over the last quarter. Write one sentence connecting these two observations."

1. Concepts: Betweenness centrality (Part 2) + Community contraction (Part 3), connected via the "low-degree-high-betweenness = critical lean broker" reading from the centrality comparison table.

2. Why they connect: A low-degree/high-betweenness node is a fragile chokepoint — its community's health is disproportionately dependent on very few ties, so member attrition (contraction) is structurally more dangerous here than in a community held together by many redundant connections.

3. Why treating them separately falls short: Looking at betweenness alone or the contraction alone misses the point the question is testing — that this course's concepts compound: a structural vulnerability (identified in Part 2) explains why an observed dynamic event (identified in Part 3) is a bigger risk than it would be elsewhere in the network.

4. Exam-ready reasoning: "Because this node's few ties are individually critical for network flow (low degree, high betweenness), the community's continued connectivity depends disproportionately on those specific ties — so losing peripheral members in this particular community is a higher structural risk than an equivalent contraction in a denser, more redundantly-connected community."